ECE501 - DIGITAL IMAGE PROCESSING

SAR-Optical Feature Fusion for Crop-Type Classification

REPORT - WEEK 7

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GitHub Repository:
https://github.com/srushtibeng-svg/DIP_2026_Project19-SOFF_Group20.git

1) Work done this week:

- Completed the inspection and analysis of SAR data availability for all four pilot fields from Germany and Uruguay.
- Selected suitable Sentinel-1 relative orbits for the four pilot fields to maintain consistency in SAR data.
- Downloaded 214 Sentinel-1 SAR acquisitions using Google Earth Engine in batches.
- Verified SAR TIFF files for image dimensions, CRS, spatial resolution, band order and acquisition metadata.

- Compared VV and VH values with Google Earth Engine for 40 sample acquisitions. All 40 comparisons passed.
- Extracted SAR time-series statistics, including mean, median, standard deviation, minimum and maximum values for VV and VH bands.
- Performed quality verification of all 214 SAR acquisitions and generated field-wise statistics CSV files.
- Generated 14 SAR time-series graphs for VV and VH backscatter values.
- Used QGIS to visualize and inspect the yield-mask candidate polygons for all four pilot fields.
- Loaded the mean yield-mask raster and corresponding candidate GeoJSON polygon layers in QGIS.
- Checked the spatial alignment between raster and polygon layers and saved separate QGIS projects for all four fields.
- Completed Sentinel-2 optical NDVI time-series processing for Germany Rapeseed, Germany Wheat, Uruguay Soybean 2021 and Uruguay Soybean 2020.
- Applied cloud masking using Sentinel-2 Scene Classification Layer (SCL) and retained observations with at least 50% valid pixels.
- Generated quality-filtered NDVI time-series CSV files and graphs for all four fields.
- Combined SAR VV/VH statistics with optical NDVI observations using a 3-day date-matching tolerance.
- Generated combined comparison graphs for NDVI, VV and VH across all four fields.
- Performed Pearson correlation analysis between NDVI and SAR VV/VH values.
- Generated eight correlation scatter plots with linear trend lines and saved the correlation summary CSV.

2) Work Planned for Next Week:

- Perform statistical significance testing using p-values for the calculated Pearson correlations.
- Review and finalize the SAR–Optical fusion results for all four pilot fields.
- Complete any remaining analysis, validation and visualizations required for the project.
- Review the QGIS projects and document the spatial analysis results and limitations.
- Prepare final comparison tables and interpret the results.
- Organize all datasets, Python scripts, CSV files, graphs and QGIS project files.

- Complete the final project report, presentation and documentation.
- Review the complete project workflow and prepare the final project submission.

3) Results:

- Successfully processed 214 Sentinel-1 SAR acquisitions across four pilot fields.
- Verified the SAR data quality and confirmed that all 214 acquisitions passed the implemented checks.
- Successfully matched 126 out of 134 optical observations with SAR observations.
- Generated 14 SAR time-series graphs.
- Generated three combined comparison graphs for NDVI, VV and VH.
- Completed Pearson correlation analysis for all four fields.

Pearson Correlation Results:

Field	NDVI vs VV	NDVI vs VH
Germany Rapeseed 2019	0.7739	0.8104
Germany Wheat 2019	-0.4391	0.1669
Uruguay Soybean 2021	0.6026	0.7103
Uruguay Soybean 2020	0.3719	0.4368

- Generated eight correlation scatter plots with linear trend lines.
- Completed QGIS visualization and alignment inspection for all four pilot fields.
- Saved the field-wise SAR statistics, NDVI time series, fusion datasets and correlation summary CSV files.

4) Challenges Faced:

- Handling a large number of Sentinel-1 SAR acquisitions and downloading them within Google Earth Engine export limits.
- Maintaining consistent SAR image dimensions, CRS, spatial resolution and VV/VH band order.
- Matching SAR and optical observations because their acquisition dates were different.

- Handling cloud-affected Sentinel-2 images and applying quality filtering before NDVI calculation.
- Verifying the spatial alignment of candidate polygons with yield-mask rasters in QGIS.
- Understanding that the generated candidate polygons represent yield-data footprints and are not independently confirmed actual field boundaries.
- Observing different correlation patterns between the four pilot fields.

5) References:

1. Google Earth Engine – Sentinel-1 GRD Dataset: https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S1_GRD
2. Google Earth Engine – Sentinel-2 SR Harmonized Dataset: https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S2_SR_HARMONIZED
3. Google Earth Engine Documentation: https://developers.google.com/earth-engine
4. QGIS Documentation: https://docs.qgis.org/
5. Rasterio Documentation: https://rasterio.readthedocs.io/
6. Pandas Documentation: https://pandas.pydata.org/docs/
7. SciPy Documentation: https://docs.scipy.org/doc/scipy/
8. NumPy Documentation: https://numpy.org/doc/